Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.058749
<D> / site : 0.176458

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Body order : 3
Clusters   : 20

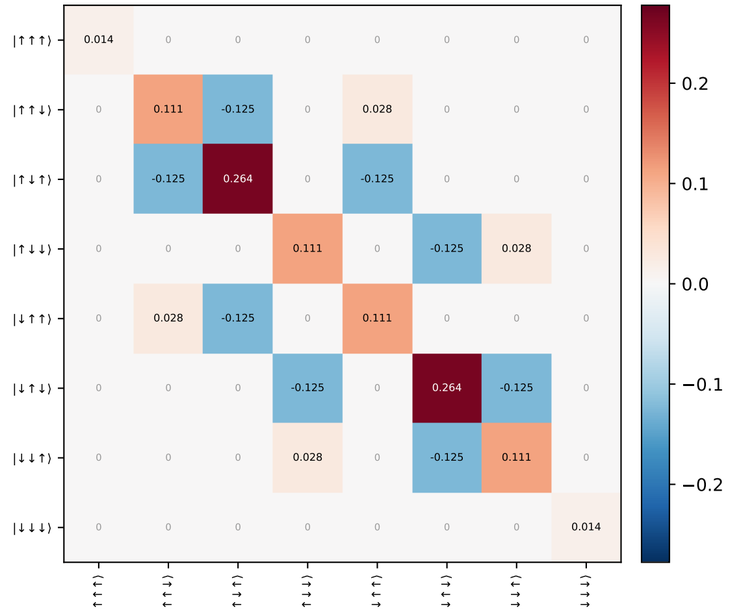
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Summary — 3-Body RDMs (Benzene, CAS(6,6))

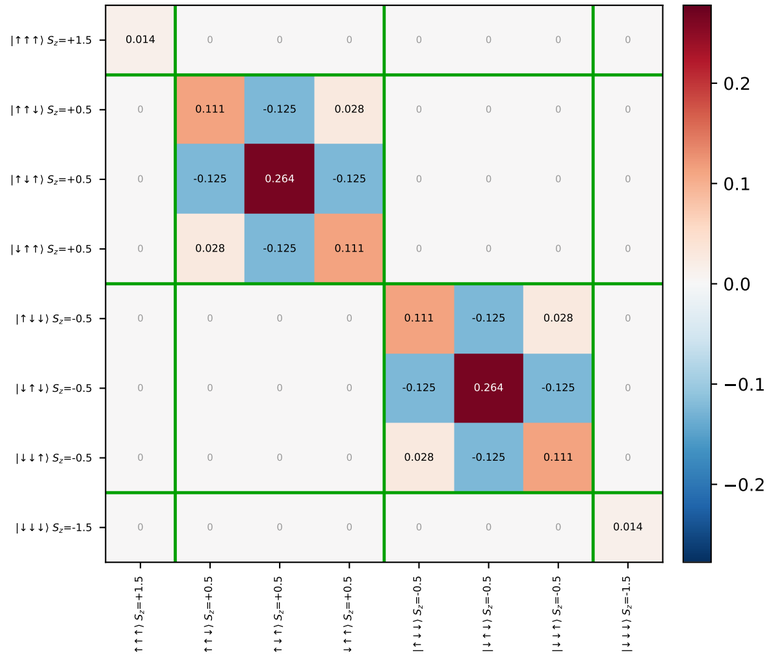
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.318468	1.381745	—	0.000000	0.000314	0.607003	MIXED
0,1,3	0.200953	1.823411	—	0.000000	0.000230	0.425910	MIXED
0,1,4	0.200953	1.823411	—	0.000000	0.000142	0.425912	MIXED
0,1,5	0.318461	1.381771	—	0.000000	0.000143	0.606998	MIXED
0,2,3	0.200951	1.823417	—	0.000000	0.000100	0.424797	MIXED
0,2,4	0.130877	2.055746	—	0.000000	0.000013	0.440588	STRONGLY MIXED
0,2,5	0.200945	1.823437	—	0.000000	0.000005	0.425921	MIXED
0,3,4	0.200950	1.823421	—	0.000000	0.000008	0.423863	MIXED
0,3,5	0.200945	1.823437	—	0.000000	0.000007	0.425921	MIXED
0,4,5	0.318461	1.381771	—	0.000000	0.000355	0.606998	MIXED
1,2,3	0.318465	1.381760	—	0.000000	0.000370	0.604583	MIXED
1,2,4	0.200949	1.823422	—	0.000000	0.000056	0.425699	MIXED
1,2,5	0.200950	1.823421	—	0.000000	0.000286	0.400235	MIXED
1,3,4	0.200949	1.823422	—	0.000000	0.000001	0.425405	MIXED
1,3,5	0.130877	2.055746	—	0.000000	0.000066	0.440587	STRONGLY MIXED
1,4,5	0.200953	1.823411	—	0.000000	0.000085	0.425912	MIXED
2,3,4	0.318465	1.381760	—	0.000000	0.000873	0.606962	MIXED
2,3,5	0.200951	1.823417	—	0.000000	0.000272	0.425931	MIXED
2,4,5	0.200953	1.823411	—	0.000000	0.000496	0.425910	MIXED
3,4,5	0.318468	1.381745	—	0.000000	0.000530	0.607003	MIXED

3-Body RDM (Sites 0,1,2) — same matrix in all three views

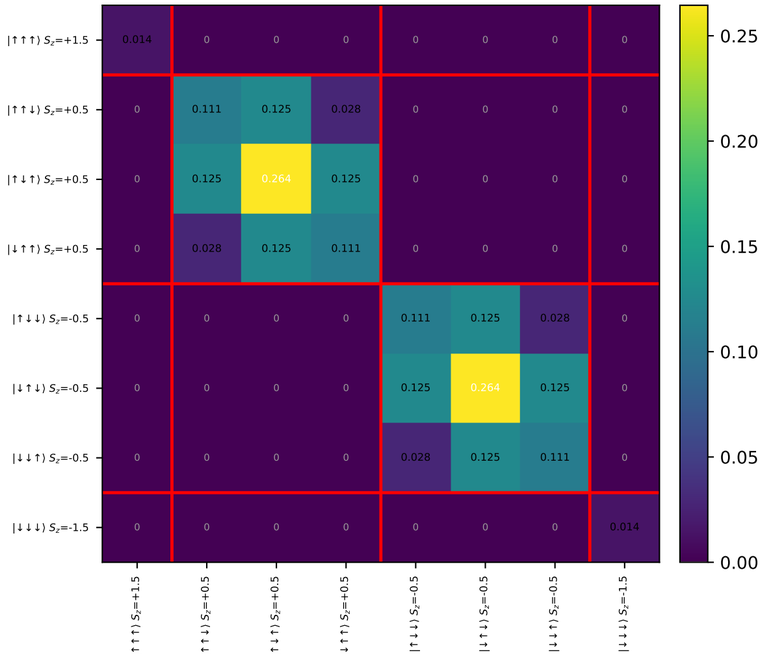
(1) Re(ρ) — index order



(2) Re(ρ) — S_z-ordered (original values)



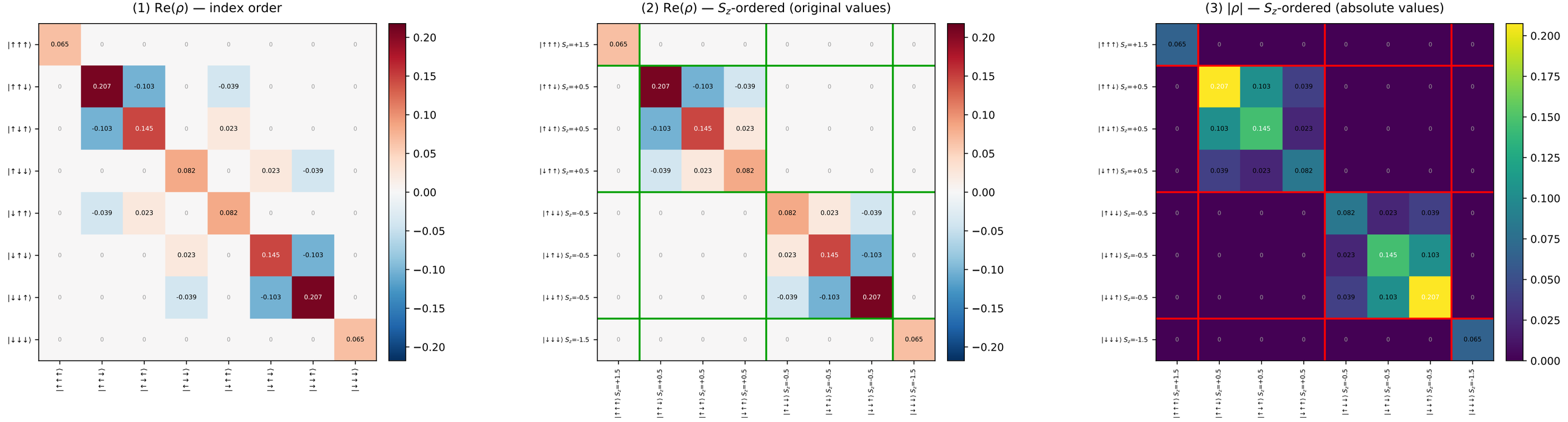
(3) $|\rho|$ — S_z-ordered (absolute values)



Trace=1.000000 Purity=0.318468 Entropy=1.381745 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

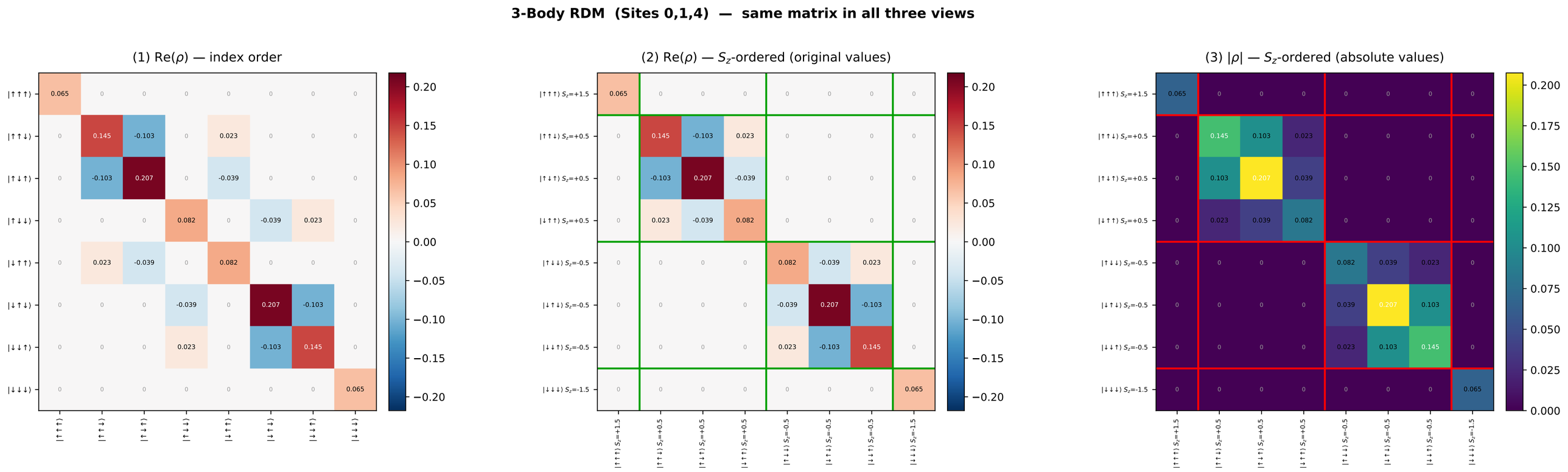
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000314 ≤ C_{UB} = 0.607003 verdict: GME

3-Body RDM (Sites 0,1,3) — same matrix in all three views

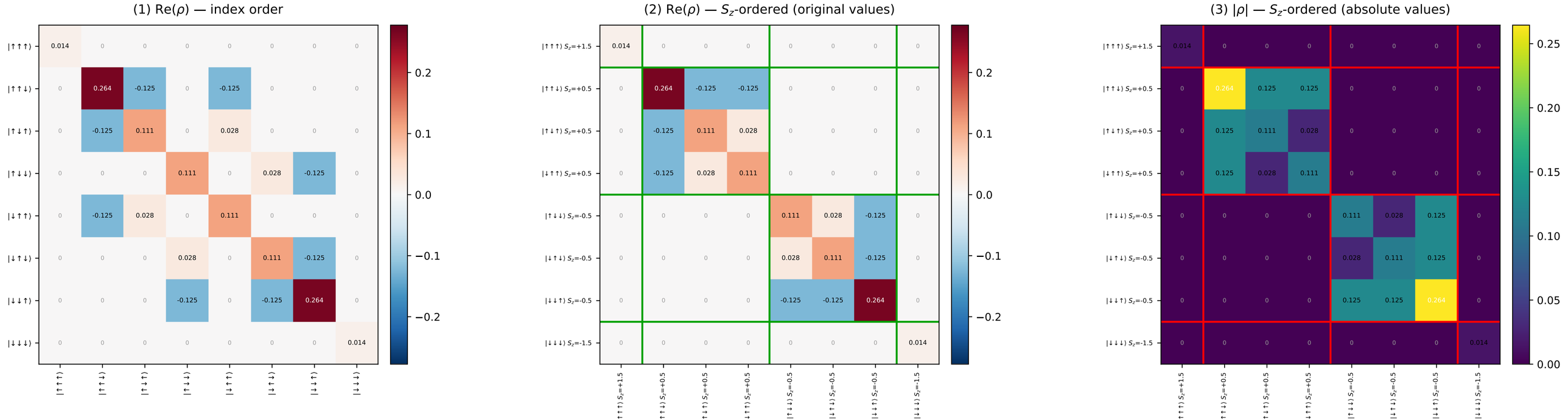


Trace=1.000000 Purity=0.200953 Entropy=1.823411 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000230 ≤ C_{UB} = 0.425910 verdict: GME


3-Body RDM (Sites 0,1,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200953 Entropy=1.823411 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000142 ≤ C_{UB} = 0.425912 verdict: GME

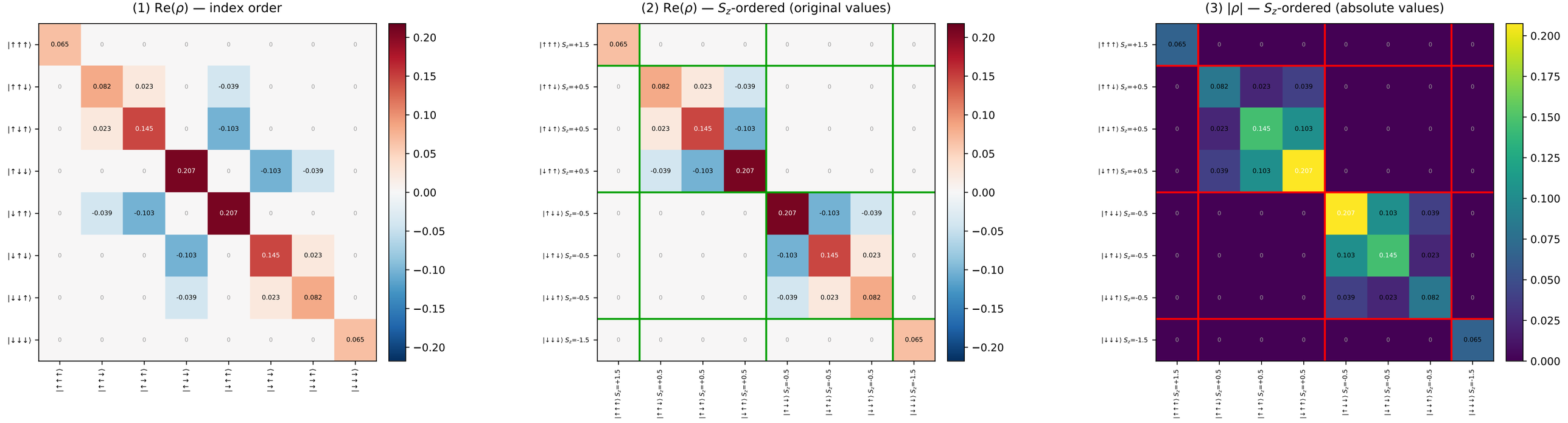
3-Body RDM (Sites 0,1,5) — same matrix in all three views



Trace=1.000000 Purity=0.318461 Entropy=1.381771 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000143 ≤ C_{UB} = 0.606998 verdict: GME

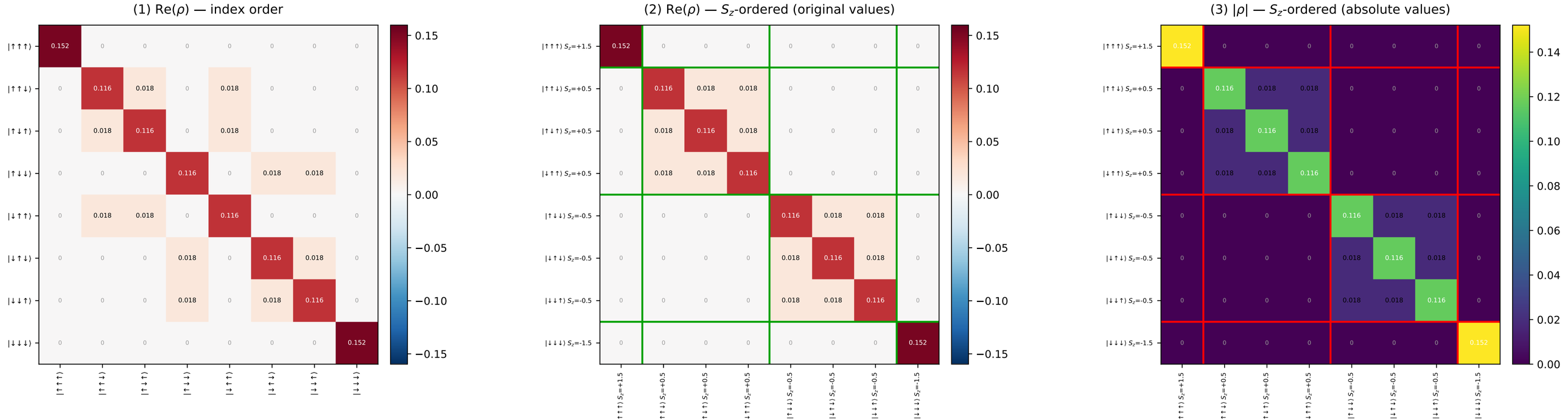
3-Body RDM (Sites 0,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.200951 Entropy=1.823417 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000100 ≤ C_{UB} = 0.424797 verdict: GME

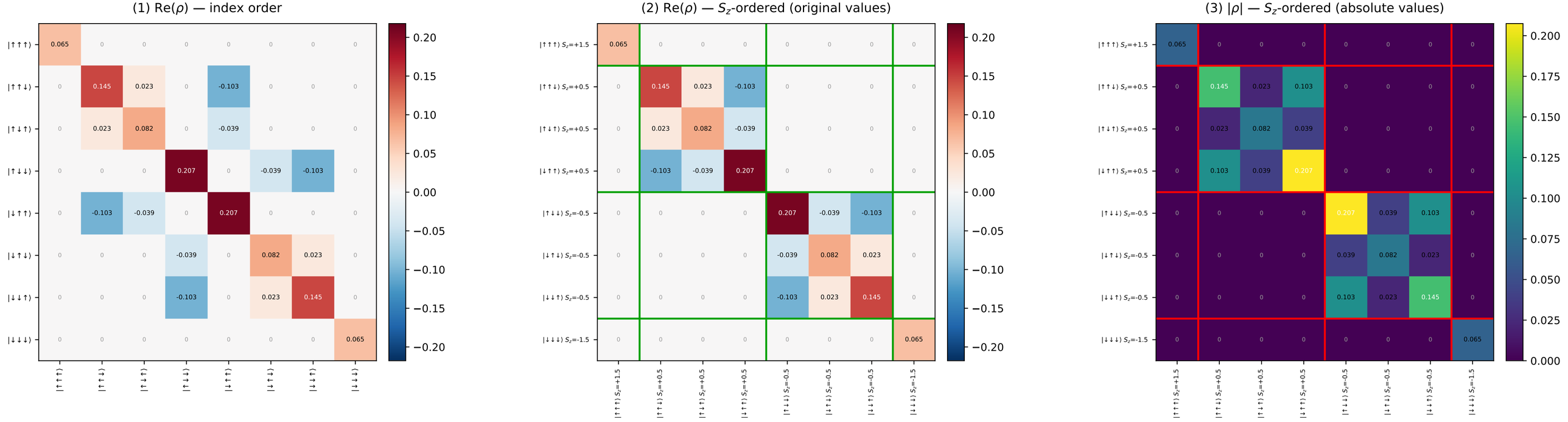
3-Body RDM (Sites 0,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.130877 Entropy=2.055746 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000013 ≤ C_{UB} = 0.440588 verdict: GME

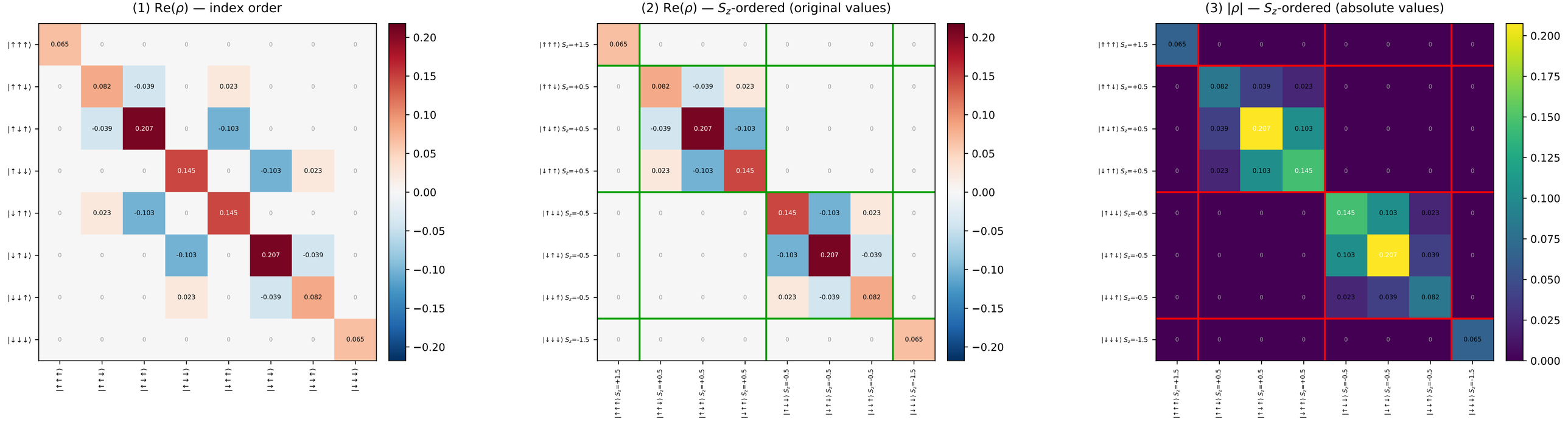
3-Body RDM (Sites 0,2,5) — same matrix in all three views



Trace=1.000000 Purity=0.200945 Entropy=1.823437 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

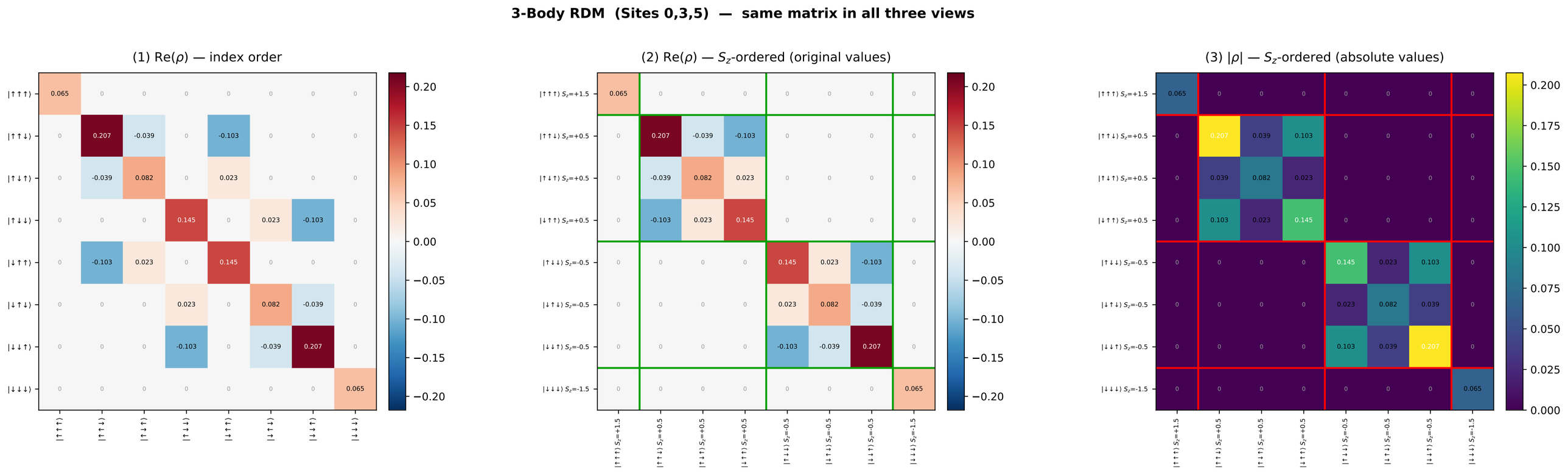
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000005 ≤ C_{UB} = 0.425921 verdict: no GME

3-Body RDM (Sites 0,3,4) — same matrix in all three views

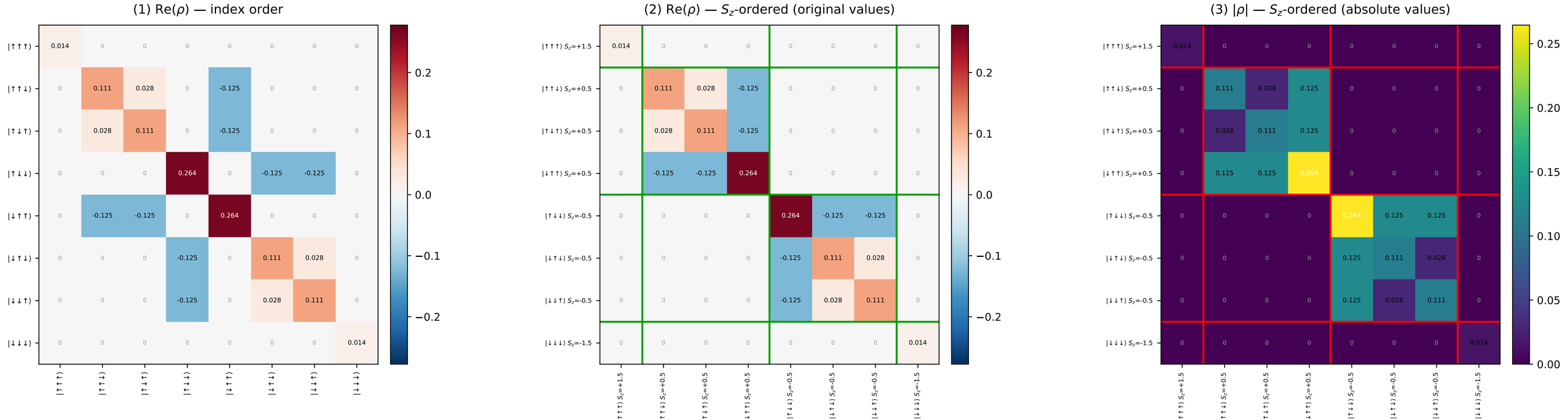


Trace=1.000000 Purity=0.200950 Entropy=1.823421 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000008 ≤ C_{UB} = 0.423863 verdict: no GME


3-Body RDM (Sites 0,3,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200945 Entropy=1.823437 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000007 ≤ C_{UB} = 0.425921 verdict: no GME

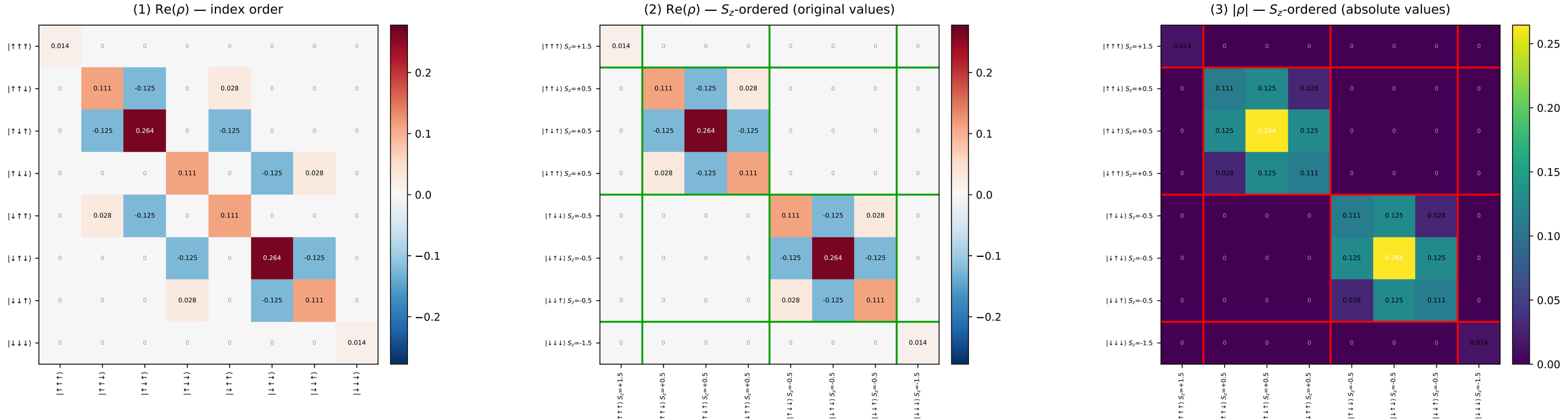
3-Body RDM (Sites 0,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.318461 Entropy=1.381771 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

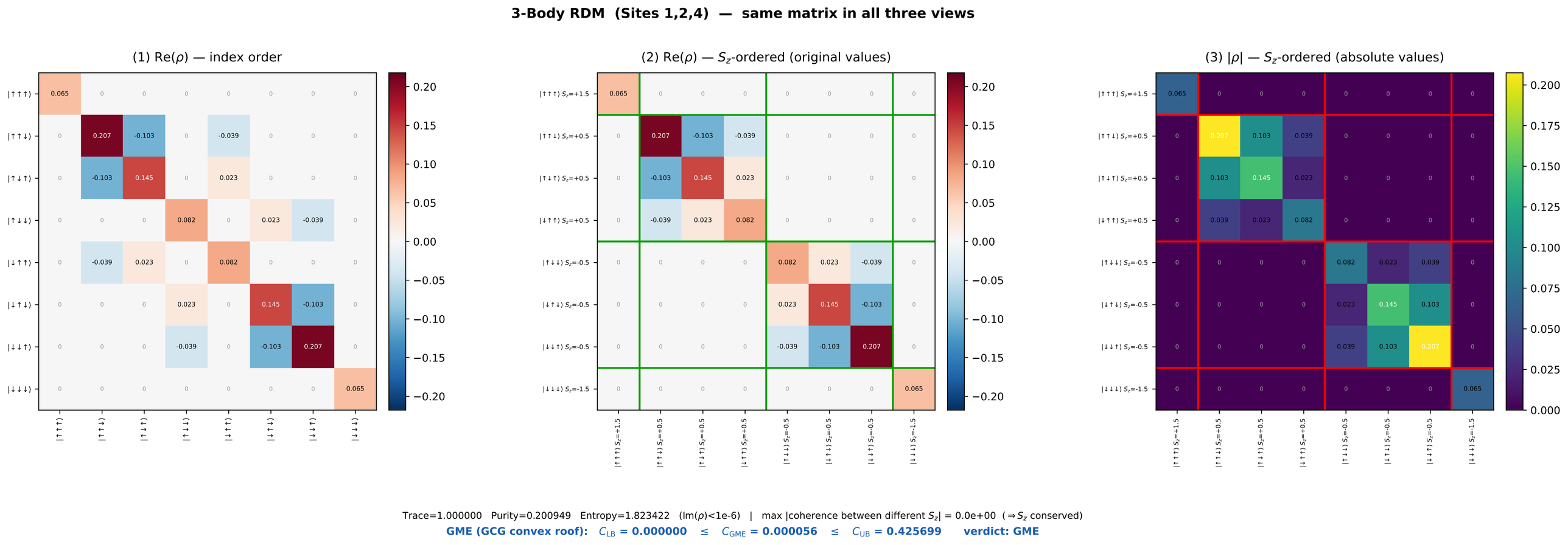
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000355 ≤ C_{UB} = 0.606998 verdict: GME

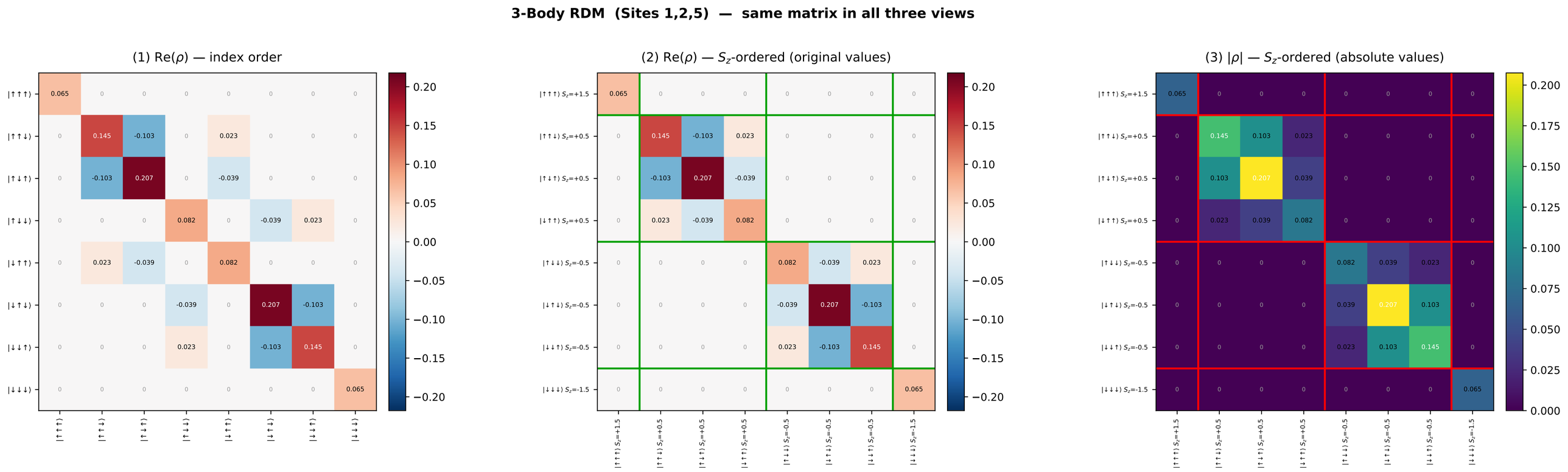
3-Body RDM (Sites 1,2,3) — same matrix in all three views

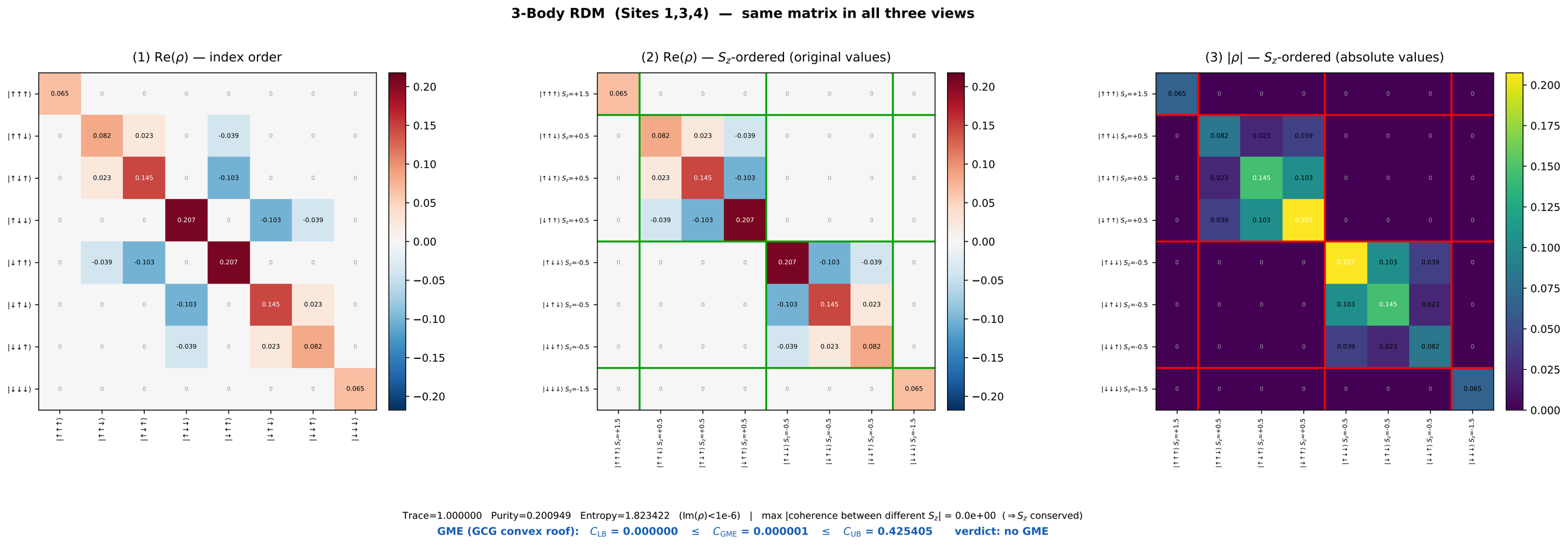


Trace=1.000000 Purity=0.318465 Entropy=1.381760 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

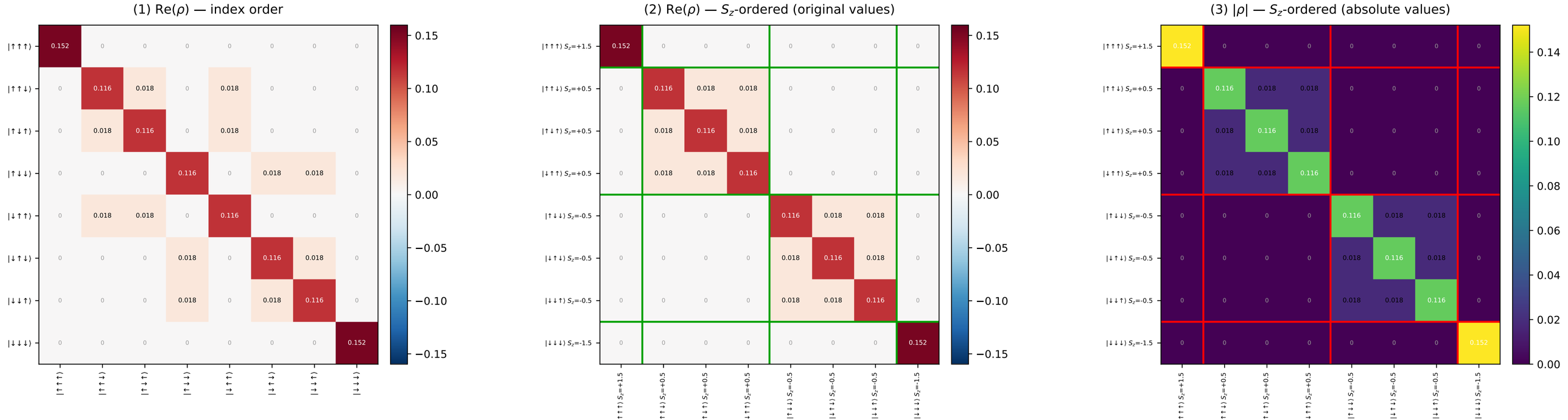
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000370 ≤ C_{UB} = 0.604583 verdict: GME


3-Body RDM (Sites 1,2,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200949 Entropy=1.823422 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000056 ≤ C_{UB} = 0.425699 verdict: GME


3-Body RDM (Sites 1,2,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
|↑↑↑⟩ S_z=+1.5
|↑↑↓⟩ S_z=+0.5
|↑↓↑⟩ S_z=+0.5
|↑↓↓⟩ S_z=-0.5
|↓↑↑⟩ S_z=+0.5
|↓↑↓⟩ S_z=-0.5
|↓↓↑⟩ S_z=-0.5
|↓↓↓⟩ S_z=-1.5
Trace=1.000000 Purity=0.200950 Entropy=1.823421 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000286 ≤ C_{UB} = 0.400235 verdict: GME

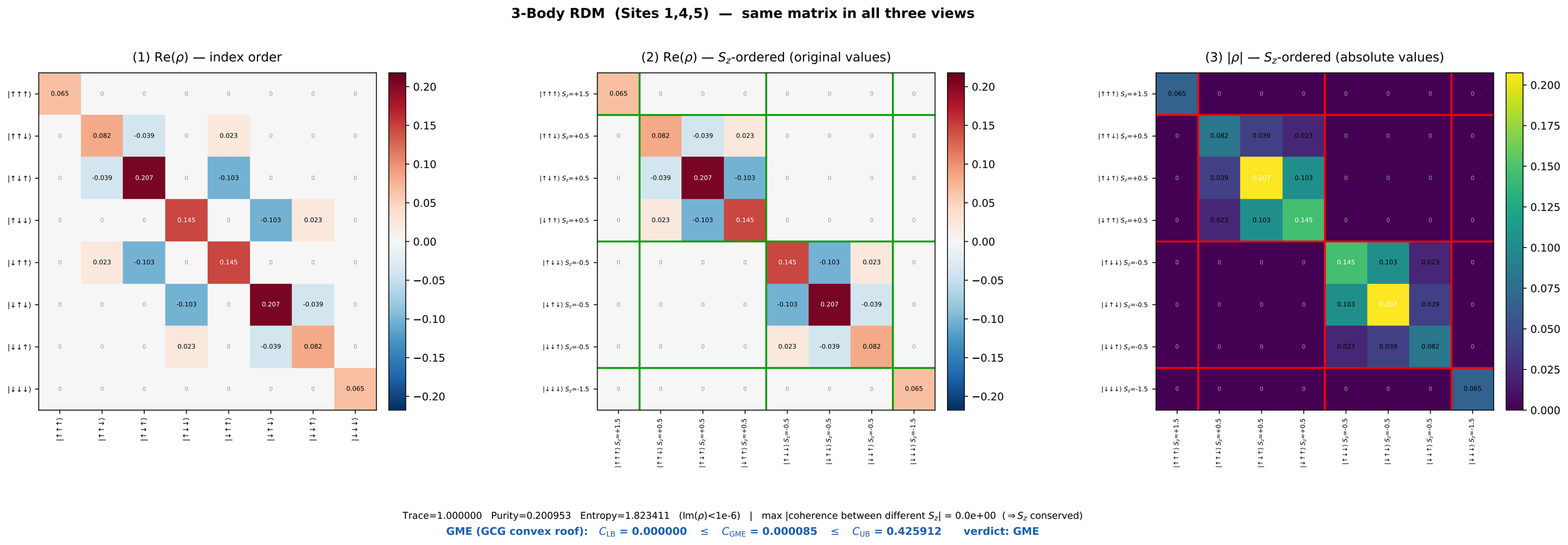

3-Body RDM (Sites 1,3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200949 Entropy=1.823422 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000001 ≤ C_{UB} = 0.425405 verdict: no GME

3-Body RDM (Sites 1,3,5) — same matrix in all three views

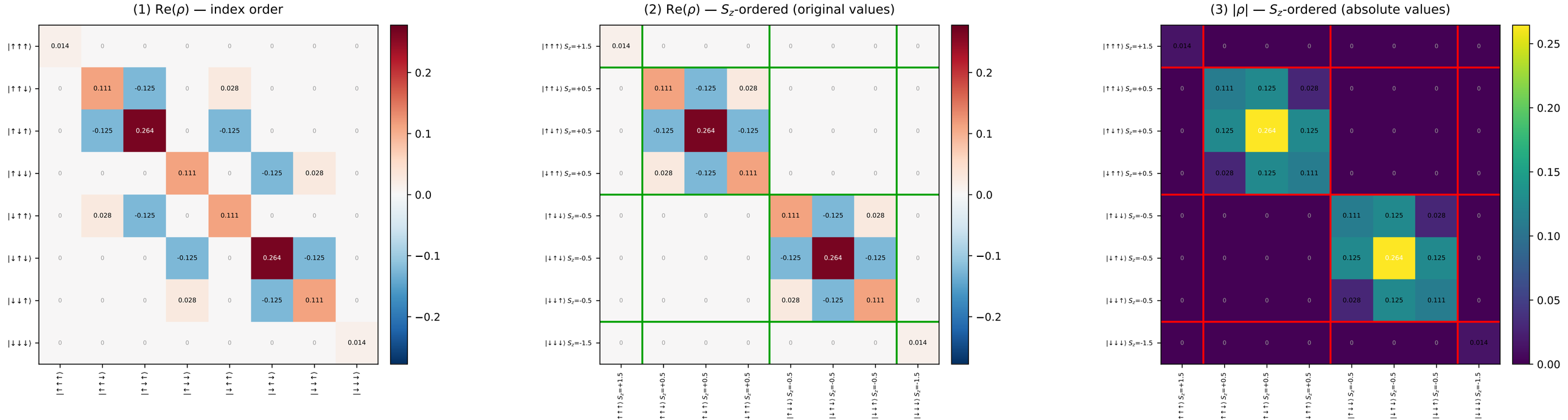


Trace=1.000000 Purity=0.130877 Entropy=2.055746 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000066 ≤ C_{UB} = 0.440587 verdict: GME


3-Body RDM (Sites 1,4,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200953 Entropy=1.823411 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000085 ≤ C_{UB} = 0.425912 verdict: GME

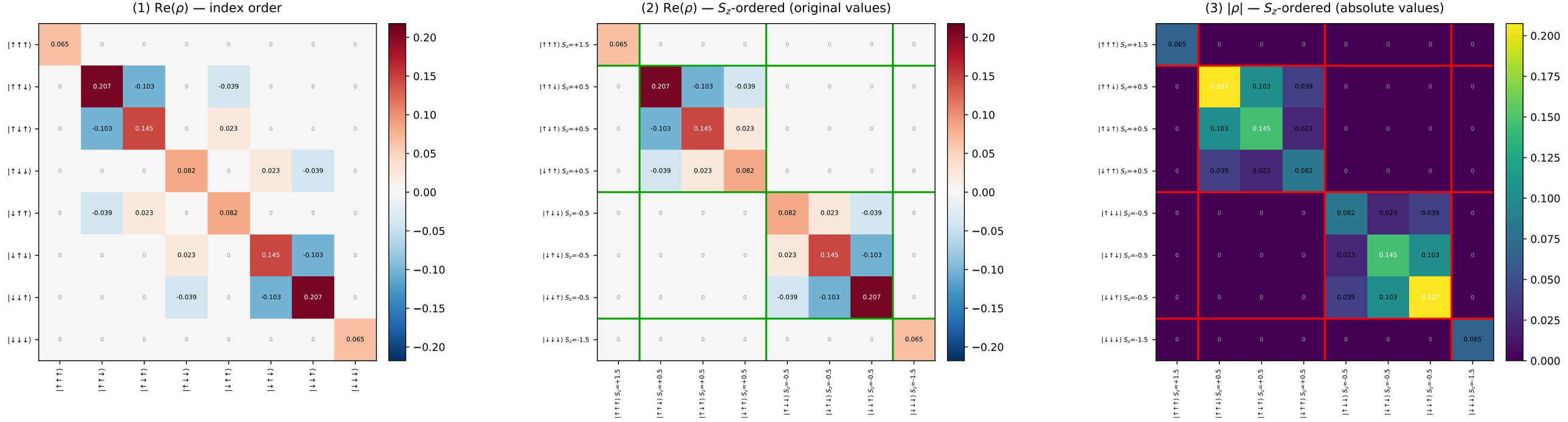
3-Body RDM (Sites 2,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.318465 Entropy=1.381760 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

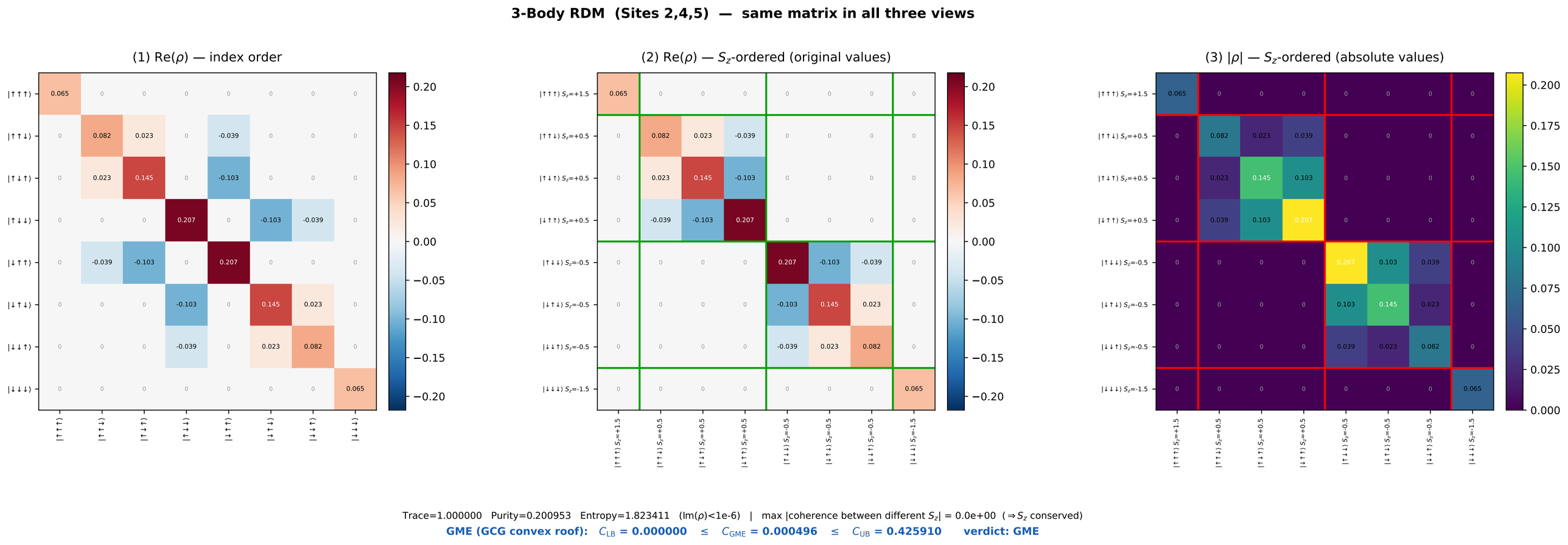
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000873 ≤ C_{UB} = 0.606962 verdict: GME

3-Body RDM (Sites 2,3,5) — same matrix in all three views

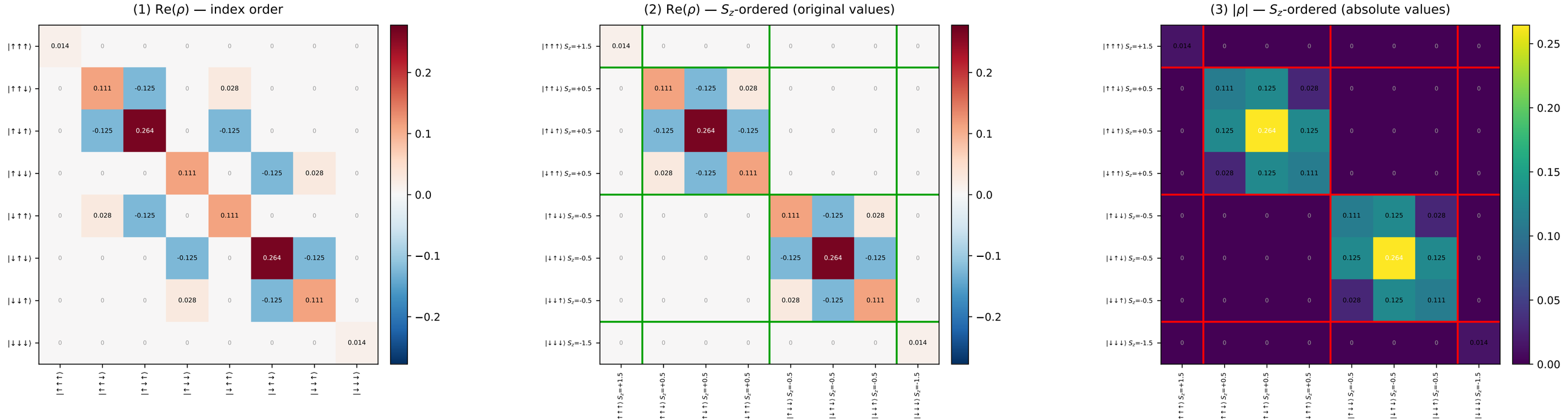


Trace=1.000000 Purity=0.200951 Entropy=1.823417 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000272 ≤ C_{UB} = 0.425931 verdict: GME


3-Body RDM (Sites 2,4,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.200953 Entropy=1.823411 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000496 ≤ C_{UB} = 0.425910 verdict: GME

3-Body RDM (Sites 3,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.318468 Entropy=1.381745 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000530 ≤ C_{UB} = 0.607003 verdict: GME